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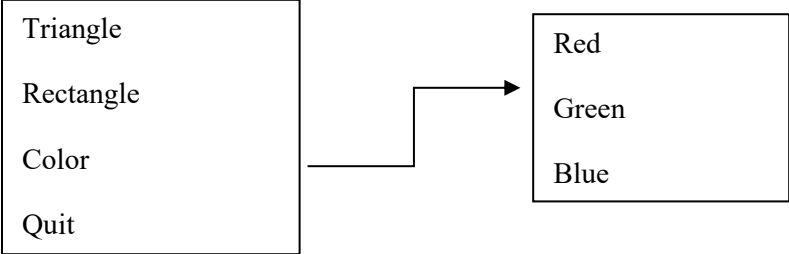
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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
VII Semester B. E. Examinations
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COMPUTER GRAPHICS AND VIRTUAL REALITY
(2022 SCHEME)
Model Paper

*Time: 03 Hours**Maximum Marks: 100**Instructions to candidates:*

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.

Q. No		PART-A	Marks
1	1.1	Write a neat diagram representing the graphics pipeline with major steps in the imaging process.	2
	1.2	List the various classes of logical input devices available to a developer of graphical applications.	2
	1.3	Describe the Three I's of Virtual Reality.	2
	1.4	Give the code segment in OpenGL to construct a triangle with the fill color as green and the boundary red colored.	2
	1.5	Differentiate between Immediate Mode and Retained mode of Computer Graphics.	2
	1.6	Give the 2D Homogenous transformation matrices for Inverse Rotation and Inverse Scaling.	2
	1.7	Write the complete taxonomy of classification of geometric planar projections.	2
	1.8	Show that transformation matrix for a reflection about a line $Y=X$ is equivalent to the reflection to X-axis followed by counter clockwise rotation of 90°	2
	1.9	Differentiate between the two modalities of Haptic feedback	2
	1.10	In 2D Geometric transformation, if Scale factors $S_x = S_y$ then the type of scaling is _____ and if $S_x \neq S_y$ then the type of Scaling is _____	2

PART-B			
2	a	Illustrate the relevance of Display list with an example code in OpenGL for Simple animated face.	5
	b	Write OpenGL program to construct 2D Sierpinski Gasket. Explain how we can extend this idea to construct 3D Gasket.	6
	c	Explain with a neat diagram the five classic components of a VR system.	5
3	a	Write an OpenGL program to demonstrate the usage of glutMouseFunc() and glutKeyboardFunc() APIs to draw a rectangle of width 8 and height 6 whenever the mouse left button is pressed or whenever the character 'r' or 'R' is pressed from keyboard.	6
	b	Any application program can obtain the measure of a device in three distinct modes, with the help of relevant examples and diagrams explain three input modes.	5
	c	What are Sound displays? Discuss the significance of different categories of Sound Displays with a suitable real-world use cases for each type.	5
OR			
4	a	Demonstrate a hierarchical menu and submenu using OpenGL functions. Attach the sub menu to the left button click. The menu and submenu should appear as shown in the figure below. Write the complete program with appropriate code (Event Handling) for each Menu entries. 	10
	b	All 3D trackers, regardless of the technology they use, share a number of very important performance parameters. Discuss the same illustrating with neat sketches.	6
5	a	Enumerate the advantages and disadvantages of Boundary-Fill and Flood-Fill algorithms. Write the C functions for both the approaches.	5
	b	Illustrate the working principle of Cohen-Hodgeman polygon clipping algorithm considering appropriate example polygon.	5
	c	Reflect the diamond shaped polygon whose vertices are $A(-1,0)$, $B(0,-2)$, $C(1,0)$ and $D(0,2)$ about the horizontal line $y=2$. Determine the new co-ordinates A', B', C' and D' on transformation. (Indicate all the steps in computation of resultant transformation). Also write the OpenGL program segment to perform this reflection (with using Inbuilt APIs for transformations)	6

		OR	
6	a	Let R be the rectangle window whose lower left hand is at L(-3,1) and upper right hand corner is at R(2,6). Apply Cohen Sutherland Line Clipping algorithm to clip the following line segments. Line 1 from A(-4,2) to B(-1,7) Line 2 from C(-1,5) to D(3,8) Line 3 from E(1,-2) to F(3,3) Line 4 from G(-4,7) to H(-2,10)	5
	b	Explain Midpoint Circle drawing algorithm by showing all the derivations for the decision parameters to scan convert a circle with center (x_c, y_c) and radius r .	5
	c	Derive the resultant composite matrix for 2D Scaling transformation with respect to any selected fixed position (x_f, y_f) indicating the sequence of steps to achieve it. Write the C function (code snippet) to achieve the same without using built-in OpenGL Library functions.	6
7	a	With the aid of neat block diagram discuss the 3-Dimensional viewing pipeline giving the functionality of all the intermediate stages	5
	b	Discuss the steps to achieve 3-D rotation of an object about an axis that is not parallel to one of the coordinate axes or rotation about an arbitrary axis with neat sketches representing all intermediate steps.	5
	c	Illustrate with neat sketches how the region code approach of 2D clipping can be extended to three-dimensional point and line clipping	6
		OR	
8	a	Design and develop an interactive graphics program using appropriate OpenGL transformation functions to perform the following operations (independently) on a 3D-Cube: • Translate the cube with key press 'T' or 't'. • Orient it with key press 'R' or 'r' • Zoom in and Zoom out with middle button. The output must be adjusted suitably when the window is resized or moved to a new position	10
	b	Differentiate between two main classes of Hidden Surface Removal techniques.	6
9	a	Write the schematic diagram illustrating the various aspects of VR modelling cycle. Briefly describe the different components involved.	8
	b	Bring out the significance of Behavior Modeling with use cases. Discuss about virtual human (or agent), group of such agents and their behavior.	8
		OR	

10	a	Discuss the features of Geometric modeling, illustrating in detail the various aspects involved. Summarize several methods by which Object surfaces can be constructed.	8
	b	Discuss the stages of Physical Modeling pipeline with appropriate mathematical formulations.	8